

lec 2:

A set S is said to be countable if there is

a one-to-one (injective) map from S to the set $\mathbb{N}$.

We have defined it as bijective in class.
 Showing an injective map alone is enough for countable set definition.

Examples:

① $\mathbb{N}$

$$f: \mathbb{N} \rightarrow \mathbb{N}$$

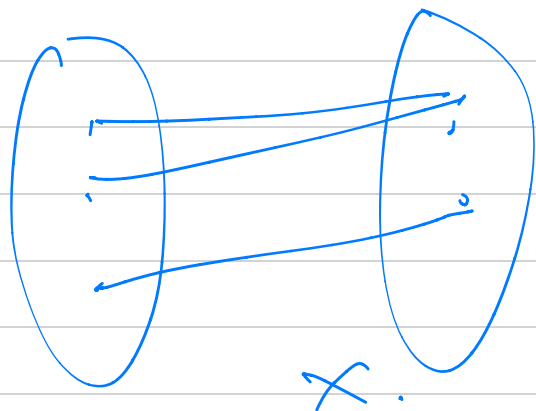
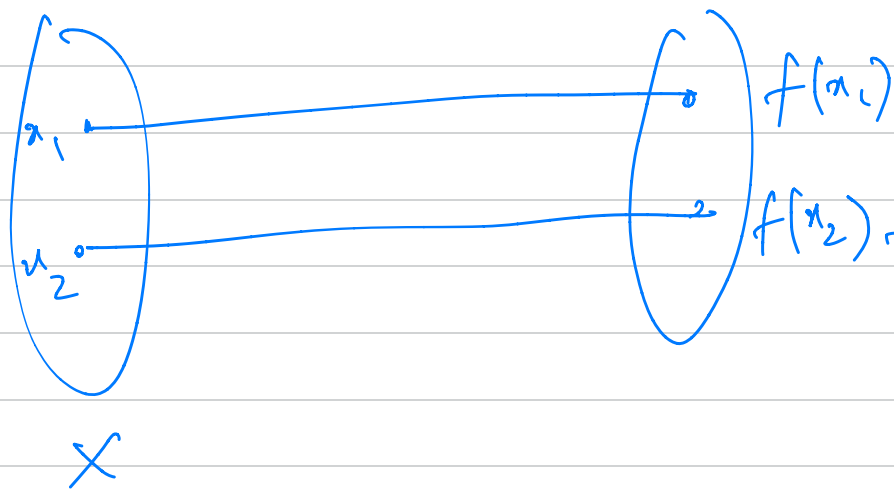
$$f(n) = n.$$

bijective map

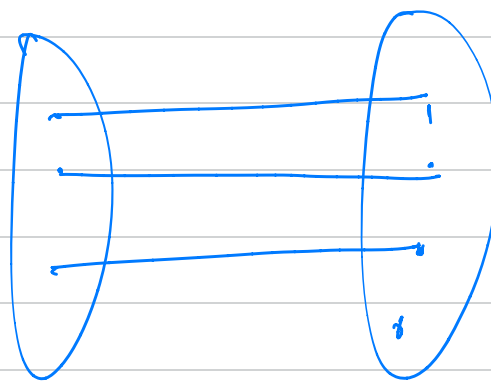
$f: X \rightarrow Y$ is said to be bijective map if it is both injective & surjective.

Injective :

If $x_1, x_2 \in X$ $x_1 \neq x_2$
 $f(x_1) \neq f(x_2)$.



X
not injective
Surjective $\checkmark$



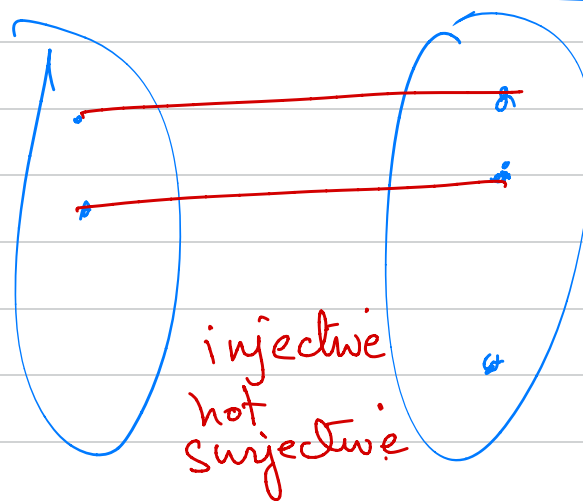
injective $\checkmark$
Surjective $\times$

Surjective (onto)

Every element in Y should have at least one x that maps to it.

$$\forall y \in Y \quad \exists x \in X \text{ st } f(x) = y.$$

If $f: X \rightarrow Y$ is a bijective map and X, Y are finite then $|X| = |Y|$.



$$\begin{aligned} &\text{Surjective} \\ \Rightarrow &|Y| \leq |X| \\ &\text{injective} \\ \Rightarrow &|Y| \geq |X| \end{aligned}$$

Countable sets

Examples:

$$\textcircled{a} \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$f: \mathbb{Z} \rightarrow \mathbb{N}$$

$$0 \rightarrow 1$$

$$1 \rightarrow 2$$

$$-1 \rightarrow 3$$

$$2 \rightarrow 4$$

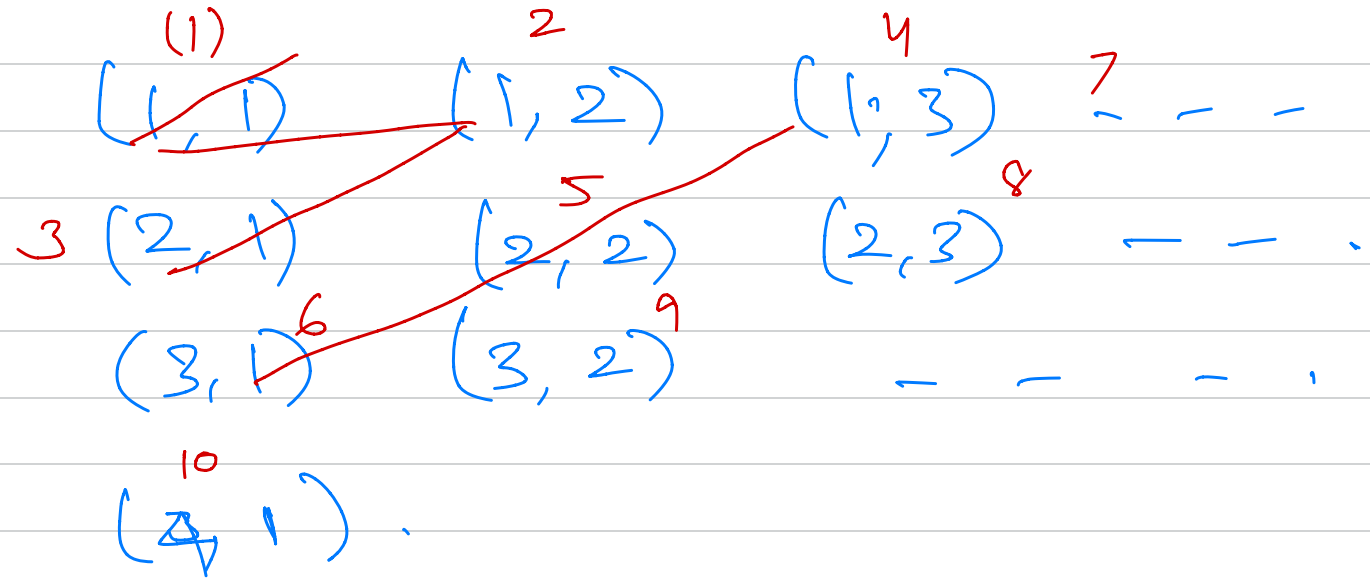
$$-2 \rightarrow 5$$

$\vdots$

Clearly
a bijective
map.

$$(c) \mathbb{N}^2 = \{ (x_1, x_2) \mid x_1, x_2 \in \mathbb{N} \}.$$

$\mathbb{N}^2$ is also countable.



(d) Suppose A is countable A^m is countable
 (follows from (c) using inductive proof) for some $m \in \{1, 2, \dots\}$

(e) $\mathbb{Q}$ rational number set is countable.

because $\mathbb{Z}$ is countable

$$\mathbb{Q} = \{ p/q \mid p, q \in \mathbb{Z}, q \neq 0 \}$$

Exercise

⑦ Subset of a countable set is always countable.

⑧ Every finite set is countable.

Uncountably infinite

If S is not countable then it is referred to as uncountable set.

⑨ $S = \{0, 1\}^*$ $\mathbb{Z}_2 = \{0, 1\}$

Including
infinite
length
sequences.

$$S = \mathbb{Z}_2 \cup \mathbb{Z}_2^2 \cup \mathbb{Z}_2^3 \dots$$

$$\begin{array}{l} 0 \\ 1 \end{array} \} \mathbb{Z}_2$$
$$\begin{array}{l} 00 \\ 01 \\ 10 \\ 11 \end{array} \} \mathbb{Z}_2^2$$
$$\begin{array}{l} 000 \\ \vdots \end{array} \} \mathbb{Z}_2^3$$

Claim: $S = \{0,1\}^*$ is uncountable.

Suppose not then S can be written as

$$S = \{\underline{x}_1, \underline{x}_2, \underline{x}_3, \dots\}.$$

Cantor's
diagonalization
technique

$\underline{x}_1$: 0 1 1 0 0 1 1 0

$\underline{x}_2$: 0 0 1 0 0 0 1 0 0 1 - -

$\underline{x}_3$: 1 0 1 0 0 1 1 0 1 1 - -

$\vdots$

$\underline{y}$ = 1 1 0 - - -

We'll come up with $\underline{y} \in S$ st $\underline{y} \neq \underline{x}_n$
 $\forall n \in \mathbb{N}$.

$$\underline{y} = (\overline{x_1(1)} \quad \overline{x_2(2)} \quad \overline{x_3(3)} \quad - \quad - \quad - \quad \dots)$$

Clearly $\underline{y} \neq \underline{x_n} \quad \forall n \in \mathbb{N}$.

$$y(n) = \overline{x_n(n)} \neq x_n(n) \Rightarrow \underline{y} \neq \underline{x_n}$$

⑥ $[0, 1]$ is an uncountable set.

any $x \in [0, 1]$ can be represented
using $\hat{x} \in \{0, 1\}^*$

$$\begin{array}{ll} 0.5 & \longrightarrow 0.1 \\ 0.25 & \longrightarrow 0.01 \\ 0.125 & \longrightarrow 0.001 \end{array}$$

$$\frac{1}{2} + \frac{1}{2^2} + \dots = 1$$

There is
bijective map
from $[0, 1] \longleftrightarrow \{0, 1\}^*$

$$\mapsto 0.1111\dots$$

Union of countably infinite # of sets:

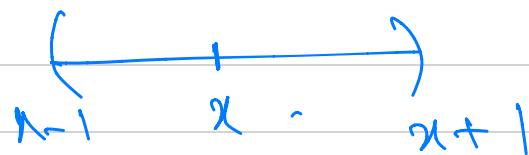
$$\bigcup_{n=1}^{\infty} A_n = \left\{ x \in \Omega \mid \exists n \in \mathbb{N}, x \in A_n \right\}$$

$$\bigcap_{n=1}^{\infty} A_n = \left\{ x \in \Omega \mid x \in A_n \forall n \in \mathbb{N} \right\}$$

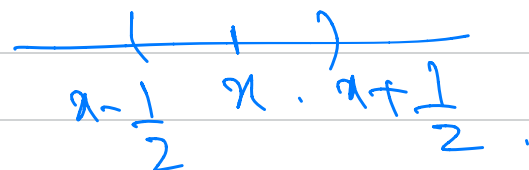
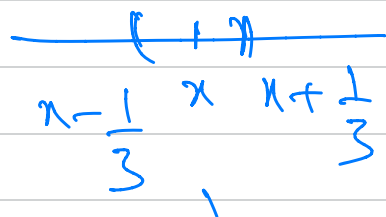
Examples:

$$\textcircled{1} \quad A_n = \left(x - \frac{1}{n}, x + \frac{1}{n} \right).$$

claim: $\bigcap_{n=1}^{\infty} A_n = \{x\}.$

A_1 

$$A_1 \supseteq A_2 \supseteq A_3 \dots$$

 A_2  A_3 

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|

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Suppose we look at $x+\epsilon$ where $\epsilon > 0$.

$$x+\epsilon \notin \bigcap_{n=1}^{\infty} A_n$$

for n st

$$x+\frac{1}{n} < x+\epsilon$$

For $n > \frac{1}{\epsilon}$.

$$x + \epsilon \notin A_n.$$

Similarly $x - \epsilon \notin A_n$ if $n > \frac{1}{\epsilon}$.

Therefore x is the only element in $\bigcap_{n=1}^{\infty} A_n$.

② $A_n = \left[0, x - \frac{1}{n}\right).$

Exercise

Show that $\bigcup_{n=1}^{\infty} A_n = [0, x).$

$$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}.$$

$$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}.$$

Event Space: $\mathcal{F}$ needs to be a σ -algebra.

① $\emptyset \in \mathcal{F}$

② If $A \in \mathcal{F}$, $A^c \in \mathcal{F}$

③ $A_1, A_2, \dots \in \mathcal{F}$, $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Derived properties:

① $\Omega \in \mathcal{F}$

as $\emptyset \in \mathcal{F}$
 $\emptyset^c = \Omega \in \mathcal{F}.$

$$\textcircled{2} \quad A_1, A_2, \dots \in \mathcal{F}$$

$$\Rightarrow \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}.$$

$$\Rightarrow A_1^c, A_2^c, \dots \in \mathcal{F} \quad (\text{property 2})$$

$$\Rightarrow \bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F} \quad (1, 3).$$

$$\Rightarrow \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c \in \mathcal{F} \quad (1, 2)$$

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$$\bigcap_{n=1}^{\infty} A_n$$

Axioms of Probability

$$P : \mathcal{F} \rightarrow [0, 1].$$

Non-negativity (a)

$$\forall A \in \mathcal{F}, \quad P(A) \geq 0$$

Normalization (b)

$$P(\Omega) = 1$$

Additive (c)

If A, B are disjoint i.e., $A \cap B = \emptyset$

$$P(A \cup B) = P(A) + P(B).$$

$$\left\{ \begin{array}{l} \text{If } A_1, A_2, A_3, \dots \text{ that are} \\ \text{pairwise disjoint } A_i \cap A_j = \emptyset \\ \forall i \neq j \\ P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \end{array} \right.$$

Derived properties

$$\textcircled{1} \quad P(\phi) = 0.$$

$$A \in \mathcal{F}, \quad A \cap \phi = \phi.$$

$$P(A) = P(A \cup \phi) \stackrel{\text{additivity}}{=} P(A) + P(\phi).$$

$$P(\phi) = P(A) - P(A) = 0.$$

$$\textcircled{2} \quad P(A^c) = 1 - P(A).$$

$$A \cap A^c = \phi \quad \left| \begin{array}{l} \text{normalization} \\ 1 = P(\Omega) = P(A \cup A^c) \end{array} \right.$$

$$A \cup A^c = \Omega$$

$$\stackrel{\text{additivity}}{=} P(A) + P(A^c)$$

$$\Rightarrow P(A^c) = 1 - P(A).$$